

Simultaneous equations and inequalities - Set 2 - Solutions

Specification MM1.4, MM1.5, M4.16, M4.17.

Answer key

Question	Answer	Topic
1	B	Simultaneous equations and inequalities
2	A	Simultaneous equations and inequalities
3	H	Simultaneous equations and inequalities
4	E	Simultaneous equations and inequalities
5	F	Simultaneous equations and inequalities
6	C	Simultaneous equations and inequalities
7	B	Simultaneous equations and inequalities

Worked solutions

Question 1 SIM-08 · Simultaneous equations and inequalities · spec MM1.5

Which of the following inequalities has exactly the same solution set as $\frac{1}{x} > 2$?

Answer: B

If $x > 0$: $1 > 2x$, so $0 < x < \frac{1}{2}$. If $x < 0$: $\frac{1}{x}$ is negative and cannot exceed 2. So $0 < x < \frac{1}{2}$. (Multiplying by x without considering its sign is the classic error.)

Question 2 SIM-10 · Simultaneous equations and inequalities · spec MM1.5

Find the complete set of values of x for which $(x-1)(x-3)(x-5) > 0$.

Answer: A

Sign chart: the product is positive for $x > 5$, negative for $3 < x < 5$, positive for $1 < x < 3$, negative for $x < 1$.

Question 3 SIM-11 · Simultaneous equations and inequalities · spec MM1.5

Given that $|2x-3| < 5$, which of the following statements is/are true?

I $-1 < x < 4$

II $x^2 < 16$

III $x > -1$

Answer: H

$-5 < 2x-3 < 5 \Rightarrow -1 < x < 4$, so I is true and III follows from I. For II: every x in $(-1, 4)$ has $x^2 < 16$ (the largest possible $|x|$ is just under 4), so II is also true.

Question 4 SIM-13 · Simultaneous equations and inequalities · spec MM1.5

How many integers x satisfy $x^2 < 5x + 24$ and $x^2 > 3x$?

Answer: E

$x^2 - 5x - 24 < 0 \Rightarrow (x-8)(x+3) < 0 \Rightarrow -3 < x < 8$. $x(x-3) > 0 \Rightarrow x < 0$ or $x > 3$. Integers: $-2, -1$ and $4, 5, 6, 7$: six.

Question 5 SIM-15 · Simultaneous equations and inequalities · spec MM1.5

The solution set of $ax^2 + bx + c < 0$ (with $a > 0$) is $-2 < x < 5$. What is the solution set of $ax^2 - bx + c < 0$?

Answer: F

Replacing x by $-x$ reflects the graph in the y -axis, so the roots -2 and 5 become 2 and -5 : $-5 < x < 2$.

Question 6 SIM-16 · Simultaneous equations and inequalities · spec MM1.5

Find the complete set of values of x for which $\frac{x-2}{x+1} \leq 0$.

Answer: C

A fraction is ≤ 0 when numerator and denominator have opposite signs (or the numerator is 0): $x-2 \leq 0$ and $x+1 > 0$ gives $-1 < x \leq 2$; $x-2 \geq 0$ and $x+1 < 0$ is impossible. $x = -1$ is excluded as the fraction is undefined.

Question 7 SIM-18 · Simultaneous equations and inequalities · spec MM1.5

Consider the statement: for all real x , if $x > 2$ then $x^2 - 3x + 2 > 0$. Which of the following is true?

Answer: B

$x^2 - 3x + 2 = (x - 1)(x - 2) > 0$ exactly when $x < 1$ or $x > 2$. So $x > 2$ does imply it (true), but the converse fails, e.g. $x = 0$ gives $2 > 0$ without $x > 2$.